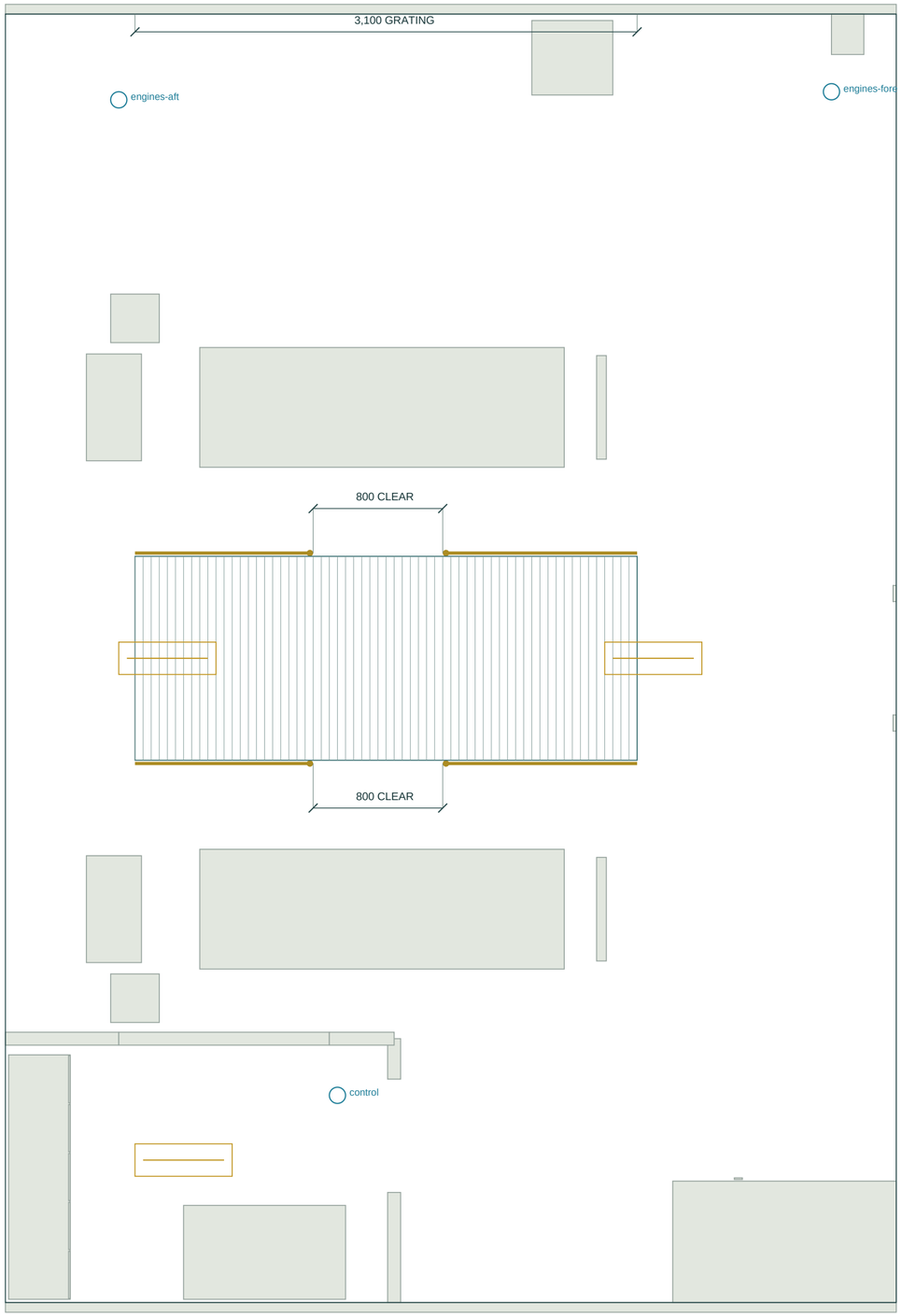
# MINATO MARU / ENGINE ROOM H1

**JS127-ER-H1-03 | CAD implementation amendment | 31 October 2026**

Development geometry study. Source positions retained; dimensions in millimetres. Plan 1:25 at A2.

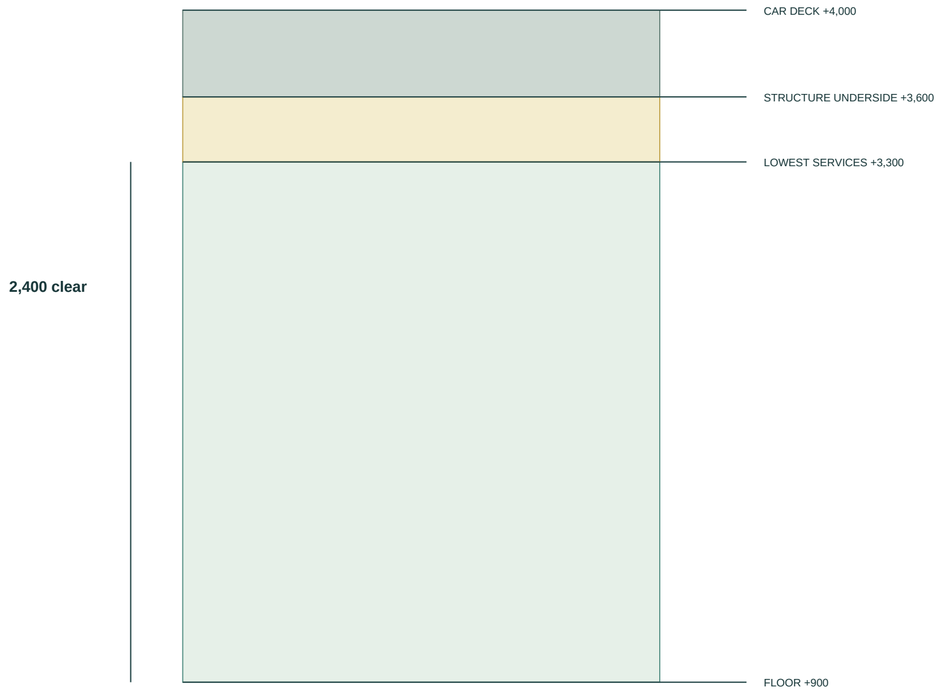



**PLAN / machinery floor +900 | port up | X from transom**

Grating 3,100 x 1,260; 6 mm bars at 50 mm pitch; top +900.

Shafts, engines, gearboxes and tank-top levels retained.

## HEIGHT AND SERVICE ACCESS / 1:20




## CHECKED IMPLEMENTATION DETAILS

Rail posts: X10,380 and X11,220; diameter 40. Inner faces leave 800 clear.

Both inboard engine rails have the service opening. Outside rails remain.

Doors X10,900 / 14,000 / 19,000 / 22,500: 2,100 high; widths 700/800/750/700.

Five new luminaire reservations: 600 x 200 x 50; underside +3,400.

Four retained fittings: two 300-square main-room; two 1,200 x 240 electrical.

Fixture centres and all eight fixed cameras are in design-h1.json and DXF.

All fixed lens centres: finite and clear by radius 120 against source boxes.

Working corridor: 1,260 between 40 mm inboard rails; floor top +900.

Clear headroom describes the working routes, not the occupied engine volume.

Illustrative pipe previews are omitted; complete service routing remains open.

400 mm deck zone is a reservation, not calculated structural scantlings.

Structure, stability, berth, fire volume, emergency escape and OEM lifts: OPEN.

Blender and World use shared geometry. Meshy pass pending human verification.

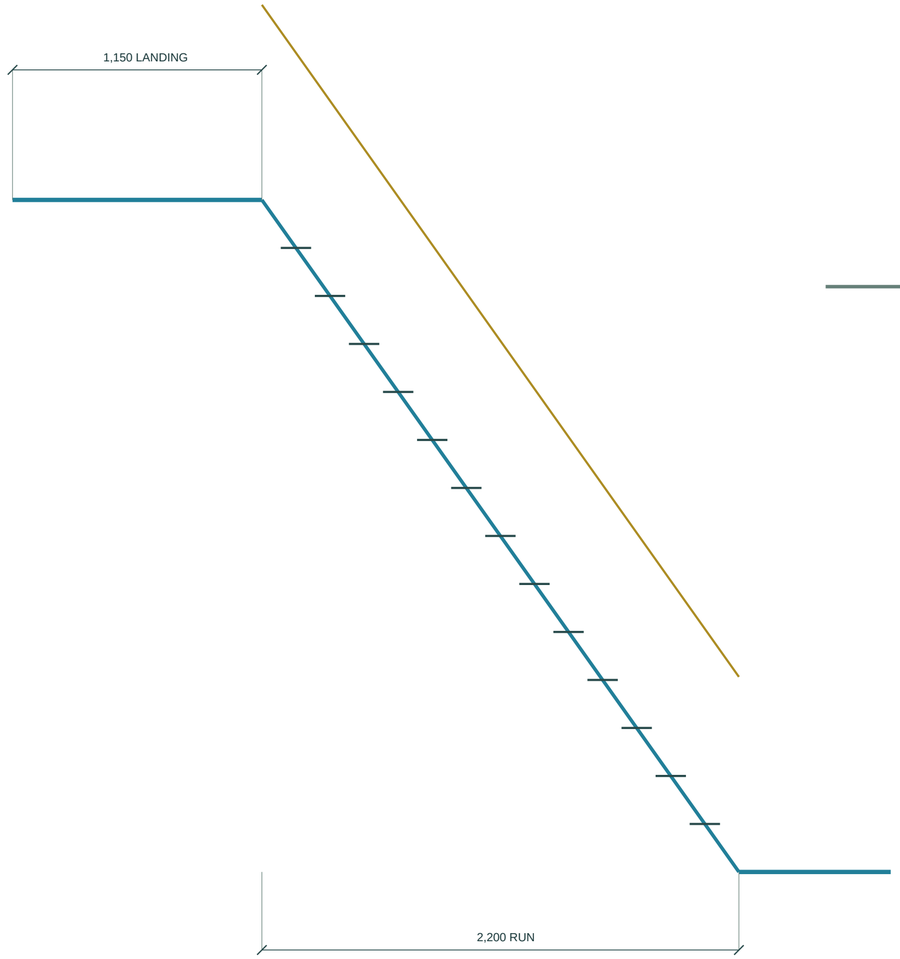
**CAD audit: mm units; native dimensions; editable layers; model comparison.**

H1 is an opt-in development study. No main promotion or fabrication approval.

# MINATO MARU / H1 LADDER AND TOP LANDING

**JS127-ER-H1-04 | Corrected access detail | 1:20 at A2 | mm**

The longer ladder requires a landing behind its head. Relocate the casing P side door; do not enter mid-flight.




**LONGITUDINAL SECTION / X forward from transom; Z above keel**

Main access only. No protected escape or lifting-route acceptance is implied.

**Development geometry study; not fabrication or regulatory approval.**

## ACCESS AUDIT / CLOSED FOR THE PROTOTYPE

New side-door opening X5,300-6,400; clear longitudinal opening 1,100.

Top landing X5,250-6,400: 1,150 long, above the same +4,000 car deck.

Ladder head X6,400; foot X8,600; rise 3,100; nominal angle 54.64 degrees.

14 risers at 221.43; nominal going 157.14. Flight is an inclined ladder.

Rail centres Y3,130 and Y3,970; diameter 40: 800 between inner faces.

Tread span Y3,140-3,960: 820 nominal. Hatch Y3,150-4,130: 980.

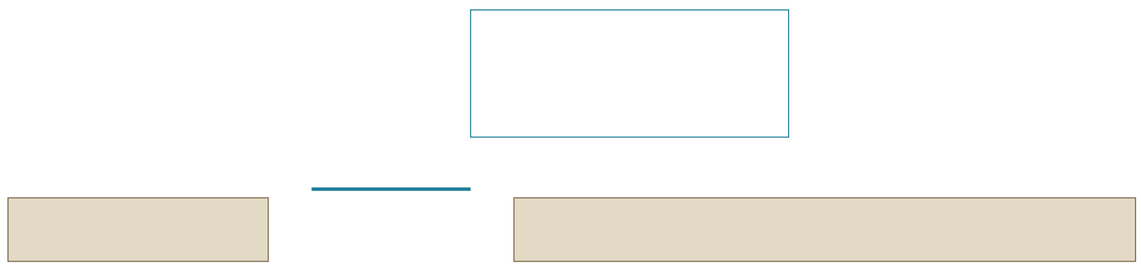
Open car-deck hatch X6,400-9,000. Ladder landing is not a second ER floor.

Walk down and back up using the new door and landing, not old X6,900.

Keep existing machinery, floor +900, tank top +700 and shaft axes.

Whole-vessel and protected emergency escape design remain open.

## PORT LOBBY / DOOR AND BENCHES / 1:30



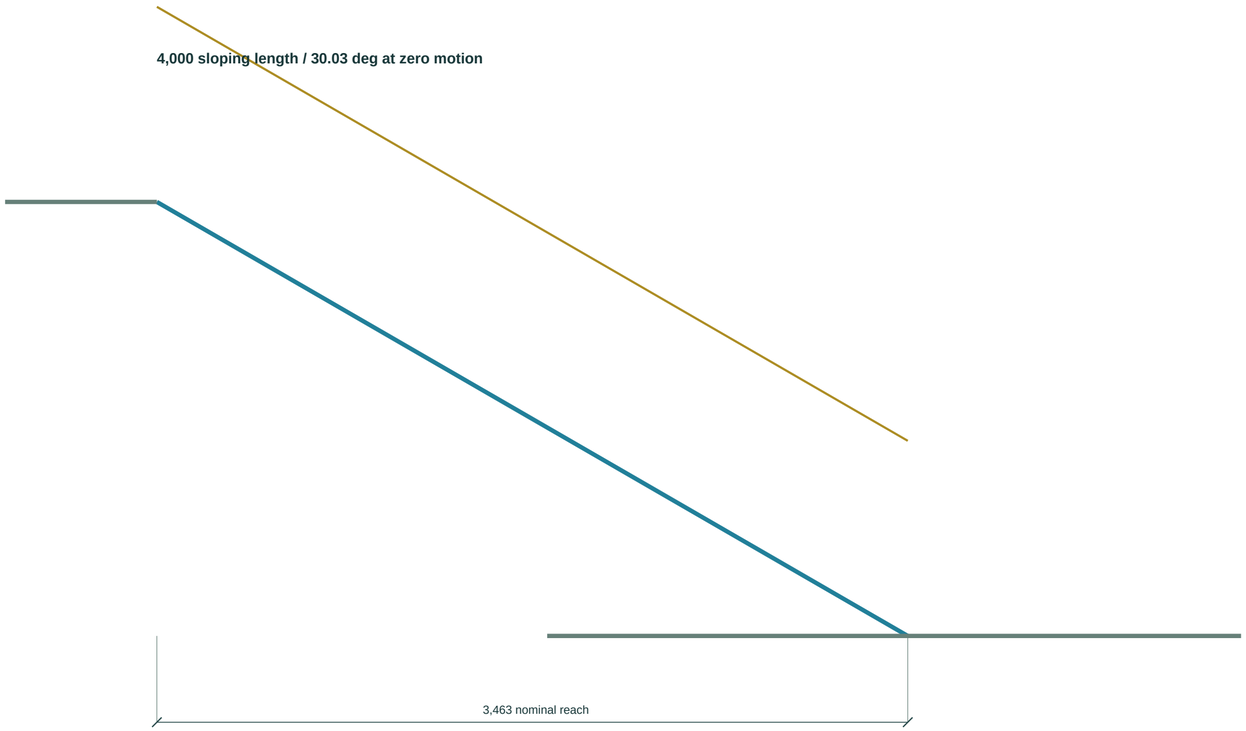
Door X5,300-6,400. Benches X3,200-5,000 / X6,700-11,000.

Retain total bench length 6,100; redistribute 1,000 forward.

# MINATO MARU / H1 BOARDING COORDINATION

**JS127-ER-H1-05 | Gangway and crew station | 1:20 at A2 | mm**

Raised vehicle deck: extend the boarding gangway rather than leave an unsupported step from the pier.




## COORDINATED GEOMETRY / LIMITS

Gangway: 4,000 long x 1,400 gross width; retained side handrails.

Fold upright under passenger deck; 4,200 clear zone reserves 200 margin.

New length maintains about 30-degree nominal boarding inclination.

Pier landing, clear pedestrian route, toe contact: verify in the walk tests.

Forward mooring station follows actual forecastle floor +6,460 above keel.

Crew post height derives from the bow-house top +60, not old 3.56 local.

Bow ramp geometry retained; both nominal berth routes require walk checks.

Hoist capacity, stow restraints, structure, tides and real access: OPEN.

No production release is authorised by this drawing.

SECTION / waterline datum uses draft 1,900; existing Minato pier top +98.

Runtime solves the actual top-surface contact under ship pitch and roll.

This is nominal game-geometry coordination; tidal and hoist design remain open.

# MINATO MARU / BLENDER ASSET FINISH

**JS127-ER-H1-06 | H1-F1 surface detail | 31 October 2026 | mm**

Retain the H1 equipment, machinery axes, access openings and 2,400 clear working envelope.

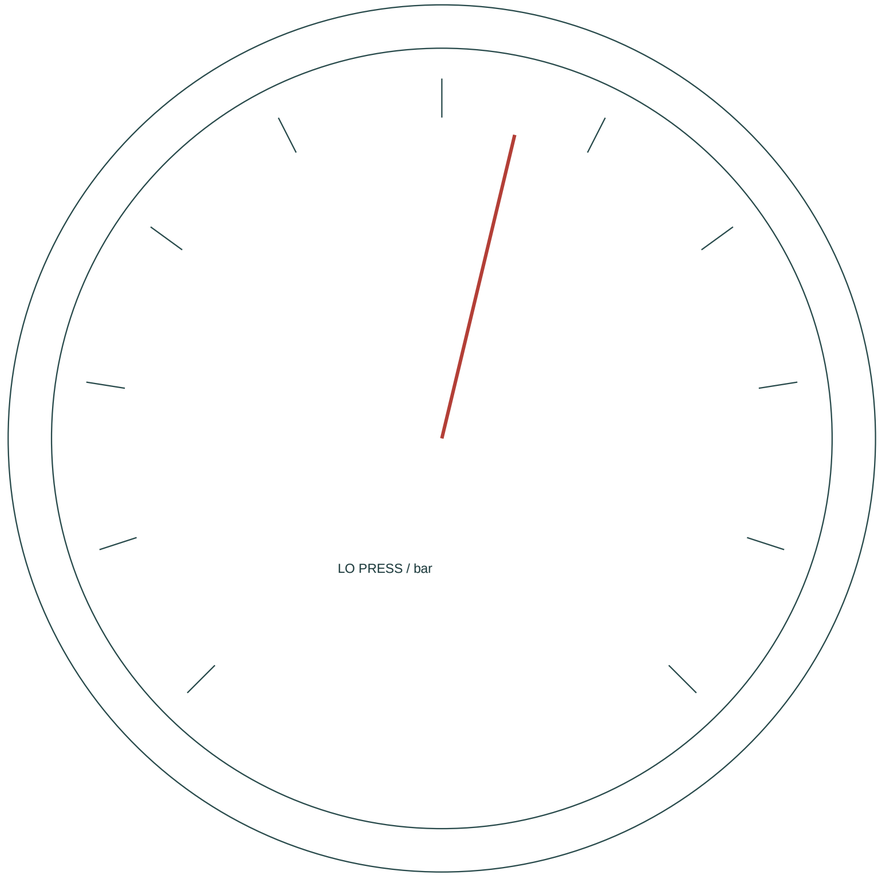
## TYPICAL EQUIPMENT EDGE / 1:2

Illustrative 600 x 400 envelope; actual part dimensions remain source-derived.

R6 MAX / inward cut; two bevel segments. Limit further to 10% of smallest part dimension.

Exclude floors, bulkheads, deckhead, partitions, sides, trunks and casings.

## DIAL FACE / 2:1




100 source dial; static source reading in Blender.

World retains its live gauges and labels.

Source colours retained; paint and bare-metal materials.

Short-range contact shade: 220 radius / 22% maximum.

Round cylinders: 24 sides; bars: 12 sides; outward faces.

No new machinery, pipe routes, fake logos or dirt decals.

Blender only. Owner supersedes the former Meshy step.